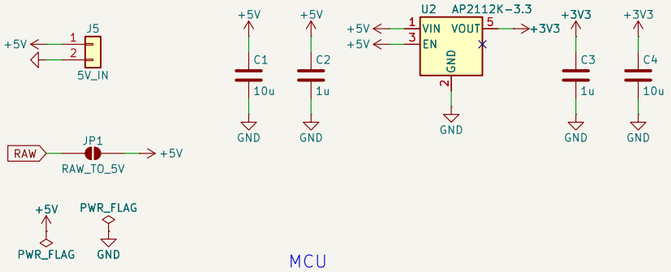
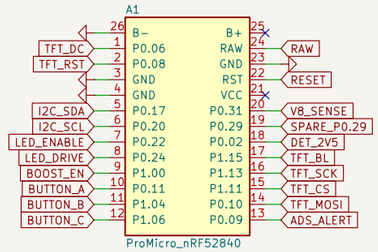
POWER

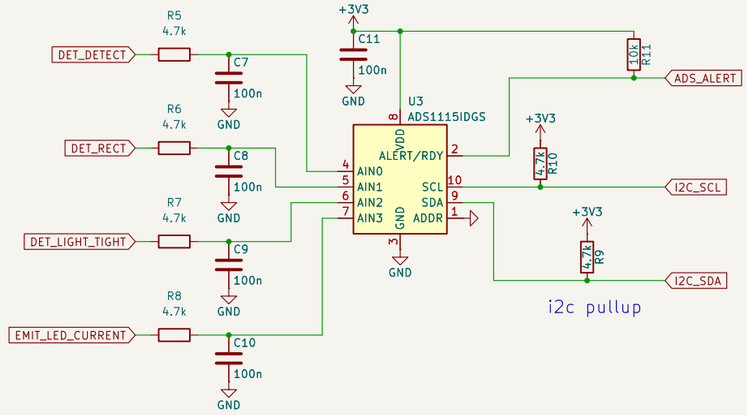


MCU

Pro Micro nRF52840 (SuperMini). CircuitPython names: board.P0_xx

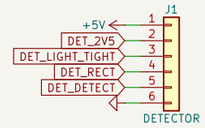


ADC: ADS1115, address 0x48 (ADDR to GND)

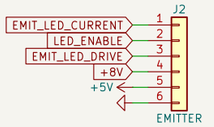


R5–R8 + C7–C10: input protection and low-pass (4.7k/100n, about 340 Hz). R9/R10: I2C pull-ups. R11: ALERT/RDY is open-drain. C11 at U3 VDD.

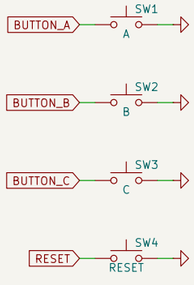
DETECTOR (JST PH 6)



EMITTER (JST PH 6)

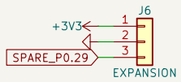


BUTTONS



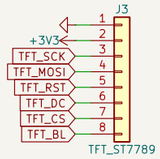
Double-press RESET for the UF2 bootloader.

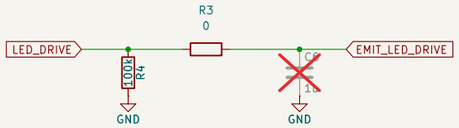
EXPANSION





DISPLAY: 2.0 in ST7789 240x320 SPI TFT, 8-pin header



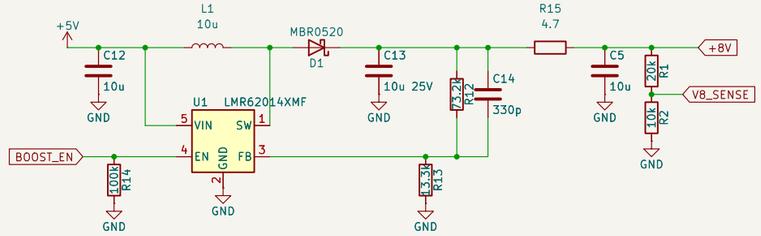


Header order GND VCC SCL SDA RES DC CS BL: check against your display. R3/C6: fit R3 = 10k and C6 to turn PWM on LED_DRIVE into a DC setpoint.

8 V BOOST FOR THE UV LED (LMR62014)

Layout: keep L1, D1, C13 and U1 pin 2 tight (short SW loop) over solid GND; R12, R13, C14 right at U1 FB.

VOUT = 1.23 V x (1 + R12/R13) = 8.0 V. C14 puts the feed-forward zero near 6 kHz. R15 + C5 filter the LED supply.



UV-Control v0.4: FIRST DRAFT. See hardware/uv-control/v0.4-design.md.

LED_ENABLE: on the emitter, HIGH turns the LED off (it grounds the current setpoint).
LOW or floating lets LED_DRIVE set the current (3.3 V -> ~26 mA).

JP1: leave open and feed 5 V into J5. RAW is most likely the Pro Micro battery pin (Rook ties RAW and B+ to its battery); close JP1 only if RAW measures about 5 V with just USB plugged in.

J1/J2 are both 6-pin JST PH: use pin-1-to-pin-1 cables and never swap them (the emitter port puts 8 V and 5 V on detector outputs).

A1 VCC (3.3 V from the Pro Micro) is unused: U2 powers the TFT and ADS1115.

U1: LMR62014 boost, 5 V -> 8 V for the UV LED. BOOST_EN high turns it on; R14 holds it off until then.

License: CERN-OHL-S-2.0
https://github.com/edgecollective/uv254

Edge Collective

Sheet: /
File: uv-control.kicad_sch

Title: UV-Control

Size: A3	Date: 2026-11-08	**Rev: v0.4**
KiCad E.D.A. 10.0.8		Id: 1/1